Lawrence Lim

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San Francisco Bay Area
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Education

University of California, Santa Barbara — Santa Barbara, CA
Doctor of Philosophy (Ph.D.) Candidate in Computer Science — Expected Graduation: June 2026

University of California, Santa Barbara — Santa Barbara, CA
Bachelor of Science (B.S.) in Computer Science (3.94 GPA) — Class of 2020
College of Engineering Honors

Work Experience

Ph.D. Candidate, Distributed Systems Lab
University of California, Santa Barbara — September 2020 – Current

- Implemented an end-to-end private AI system for large language model inference with homomorphic encryption with novel performance optimizations and matrix multiplication algorithms [1].
- Optimized key rotations by developing new protocols to eliminate race conditions while utilizing new updatable encryption primitives, resulting in an accepted research paper [5].
- Developed a virtual reality application to spatialize music, resulting in a paper at a top conference [6].
- Developed an augmented reality mobile application for spatial music, culminating in an accepted paper at a top HCI conference [7].
- Reduced workload imbalance for distributed system backends by developing a front end cache [8].
- Led research teams, forged collaborations with industry, and mentored undergraduate and graduate students, resulting in multiple successful publications.

Software Engineer Intern, Google
Sunnyvale, CA — June 2024 - September 2024

- Enabled compilation of a Python machine learning models into CKKS homomorphic encryption, leading to a 3000x performance improvement over previous infrastructure.
- Implemented optimized matrix multiplication FHE kernels, leading to a 6x performance improvement.
- Developed a pipeline in MLIR, which directly contributed to the HEIR research paper [3].

Software Engineer Intern, Oracle
Redwood City, CA — June 2021 – September 2021, June 2022 - September 2022

- Developed algorithms for optimizing for net benefit of query plan native compilation by modeling native compilation benefit and clustering to group similar query plans, leading to a patent [4].

Software Engineer Intern, Toyon Research Corporation
Goleta, CA — July 2019 – September 2020

- Improved accuracy of object detection in applications of AI models like RetinaNet and Faster R-CNN by tuning hyperparameters and implementing image transformations.
- Obtained secret security clearance, authoring progress reports for government contracts.
- Created image classification model using ResNet to determine benefits of additional LIDAR data.

Skills

Artificial Intelligence:	Large Language Models, Transformers, CNNs
Security and Privacy:	Homomorphic Encryption, Multi-Party Computation, Differential Privacy
Systems:	Databases, Distributed Systems, Compilers, Spatial Computing

Selected Publications

[1] **Lim, L.**, Kalagi, V., Agrawal, D., and El Abbadi, A. "Tricycle: Private Transformer Inference with Tricyclic Encodings" (2025)
[2] **Lim, L.**, Liu, J., Kalagi, V., Agrawal, D., and El Abbadi, A. "Hyperion: Private Token Sampling with Homomorphic Encryption" (2025)
[3] Ali, A., Choi, J., Gipson, B., Gorantala, S., Kun, J., Legiest, W., **Lim, L.**, Viand, A., Demissie, M., and Zheng, H. "HEIR: A Universal Compiler for Homomorphic Encryption" (2025)
[4] **Lim, L.**, Aghababae, L., and Swart, G. "Compiling Just the Best Bits: Finding the Maximum Benefit Shared Subexpressions in a Query Workload" (US patent 2025)
[5] **Lim, L.**, Goh, W., Agrawal, D., El Abbadi, A., Gupta, T. "QUICKeR: Quicker Updates Involving Continuous Key Rotation" (PoPETS 2024)
[6] Jiang, S., **Lim, L.**, and Sra, M. "Spatializing Music in Virtual Reality" (SUI 2023)
[7] **Lim, L.**, Goh, W., and Sra, M. "A Spatial Music Listening Experience in Augmented Reality" (UIST 2021)
[8] Zackhary, V., **Lim, L.**, Agrawal, D., and El Abbadi, A. "Cache on Track (CoT): Decentralized Elastic Caches for Cloud Environments" (EDBT 2021)